
Part V

Climbing and Age

Long-Term Effects of Intensive Rock Climbing to the Hand and Fingers

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Jens Lisse is climbing "Götterdämmerung", Frankenjura, Germany.
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13.1 Introduction

The connection between rock climbing and degenerative arthritis of the finger joint has been debated and investigated by various authors [1–4]. Our analysis includes an overview of data and patients spanning approximately 30 years [5]. Some have ended their sporting careers over the years; others—some of them over 60—are still more or less active in their sport. The risk of polyarthritis of the finger joints can be well assessed on the basis of these cases. There is certainly a correlation between the number of years of climbing, the level of climbing, and a higher incidence of osteoarthrotic changes in the finger joints [1, 2, 5–10]. In addition to radiographic examinations, however, the symptoms and clinical signs are also decisive for a further prognosis for the climber.

13.1.1 Clinical Findings

Pain in the area of the middle and, to a lesser extent, in the end joints of the fingers is a common problem faced by sports climbers. These complaints mostly occur after intensive training or climbing periods. Climbers report load-dependent pain but also pain at rest, sometimes with a dull sensation of swelling [1, 11]. At times, the finger middle joints (PIP) can no longer be fully extended and show an extension deficit of 5–15°. Climbers also often describe a loss of fine motor skills [12–14]. In the morning, an unpleasant stiffness of the joints becomes noticeable, which usually improves after a few movement exercises. At the end joints, an adaptive thickening of the joints often becomes apparent, sometimes with small “nodules” as is also known from the osteoarthritis of the DIP joint (Heberden’s osteoarthritis). Nevertheless, the underlying pathology of Heberden’s osteoarthritis is different. During the examination, in addition to a slightly spindle-shaped thickening of the middle joints, elastic joint effusions and slight instabilities of the radial and ulnar ligaments at the PIP joints can sometimes be found. The frequently

found slight instability is usually caused by the joint effusions.

Pain in the dorsal part of the PIP joint requires special attention. Pressure pain combined with painful movement restrictions and a snap phenomenon at the beginning of flexion is sometimes experienced by older climbers.

Noticeable radial or ulnar instabilities—as known in severe osteoarthritis of the DIP joint—are extremely rare. The metacarpophalangeal (MCP) joints of the fingers are clinically and radiologically rarely affected. Only a radiographic examination can confirm the diagnosis of arthrosis of the PIP or DIP joints and thus differentiate between inflammation of the capsule or joint effusions.

13.2 Adaptive Osseous Reactions

In the analysis of radiographic images of sport climbers, it is important to separate bony adaptive reactions from degenerative changes in the joints [15] (Table 13.1). This distinction is not always easy and requires precise radiological diagnostics. Schöffl et al. [15, 16] used the following protocol to distinguish between adaptive osseous reactions and osteoarthrotic findings (Table 13.1). In contrast to the scale of Kellgren and Lawrence [17], subchondral sclerosis was added to the group of physiologic, adaptive stress reactions and was not defined as osteoarthritis [8, 15]. Furthermore, osteoarthrosis was diagnosed if a Kellgren and Lawrence grade ≥ 2 was present [2, 6, 8, 15]. Prior studies showed that reactions such as cortical hypertrophy and subchondral sclerosis in the fingers could be adaptive signs of high stress in climbing and could not be attributed to pathologic osteoarthrotic changes [14, 18, 19]. The visual guidelines given in the *Atlas of Standard Radiographs* [20] were used for comparison with normal radiograph results [2, 6, 8, 15].

One of the most frequent adaptive reactions—which can often be seen clinically—is an enlargement of the joint outlines (Fig. 13.1).

In about 45% of climbers, this broadening of the joint base and surrounding connective tissue

Table 13.1 Protocol for radiographic evaluation

Protocol for radiographic evaluation (Schöffl et al. [8, 9, 15])	
Finding	Definition
Cortical hypertrophy	Visible cortical hypertrophy at the middle point of the phalanx; length in AP view or biconvex cortical hypertrophy in lateral view
Subchondral sclerosis, increased thickness of epiphysis	Stress reaction in contradiction to Kellgren-Lawrence scale
Calcification of insertion of flexor digitorum superficialis or flexor digitorum profundus tendon	Visible calcification
Broadened proximal interphalangeal joint base	Increased radius of lateral or medial base of phalanx
Broadened distal interphalangeal joint base	Increased radius of lateral or medial base of phalanx
Early osteoarthritic reaction	Kellgren and Lawrence grade ≥ 2 (note: subchondral sclerosis defined as adaptive stress reaction) or epiphyseal fracture

is already visible after 5–10 years [14, 16]. The widening of the joint surfaces should not be confused with osteophytic structures, which show a different radiographic picture. The intensity of the osseous reactions correlates with the number of active climbing years and intensity of the sport; they become increasingly pronounced with increasing duration and intensity [5, 8]. However, the first signs of adaptations can already be observed after about 3 years of intensive climbing. A study of athletes of the German National Climbing Team from 1999 to 2004 [15] showed interesting findings. Young athletes on the German Junior Climbing Team (GJNT), starting at an average age of 14 years, were compared with a control group over 5 years and again after 11 years. Stress reactions and adaptation symptoms of the skeleton were already found in about 30% at the beginning of the longitudinal study

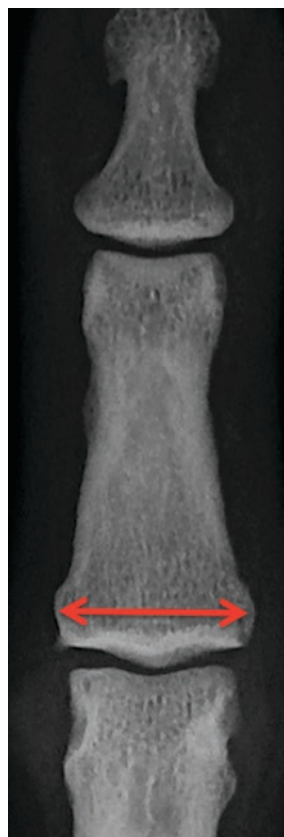


Fig. 13.1 A 48-year-old climber: climbing time over 30 years. Widening of the joint bases at the DIP and PIP joint (red arrow). Also, small mini-osteophyte at the PIP joint

[15]. In both the GJNT and the group of recreational climbers (RC), one case with an epiphyseal growth plate fracture was detected in each group. These were classified as early osteoarthrotic findings, nevertheless both had healed without any further damage in later controls [16]. Thus, they do not stand out as osteoarthrotic cases in the longitudinal analysis (Table 13.2). Slightly increased subchondral sclerosis of the cortical bone close to the joint or cortical thickening of the middle phalanges are seen frequently (Fig. 13.2). In addition, ossification of the tendon insertions (profundus and superficialis tendon) occur frequently in long-term climbers (Fig. 13.3).

Table 13.2 Radiographic changes of the German Junior National Team (GJNT), recreational climbers (RC), and the non-climbing control group (CG) (Schöffl et al. [15])

	GJNT	RC	CG
Number	19 (<i>m</i> = 15, <i>f</i> = 4)	18 (<i>m</i> = 16, <i>f</i> = 2)	12 (<i>m</i> = 8, <i>f</i> = 2)
Stress reactions	9 (47%)	5 (28%)	0
Cortical hypertrophy	5 (26%)	2 (11%)	0
Subchondral sclerosis, increased thickness of epiphysis	9 (47%)	1 (6%)	0
Calcification of insertion of flex. sup. or flex. prof. tendon	0	0	0
Broadened joint base PIP	8 (42%)	5 (28%)	0
Broadened joint base DIP	3 (16%)	0	0
Early osteoarthrotic reactions	1 (5%)	1 (6%)	0
Osteophytes PIP	0	0	0
Osteophytes DIP	0	0	0
Decreased jointspace	0	0	0
Subchondral cysts	0	0	0
			0
Epiphyseal fractures (stress fractures)	1	1	0
Calculation			
Barnett-Nordin index: (cortical thickness of the middle phalanx/ total osseous thickness of the middle phalanx (cortical and cancellous bone))	0.49 ± 0.05 (mean ± sd)	0.49 ± 0.07 (mean ± sd)	0.48 ± 0.08 (mean ± sd)

First number = *n*; second number = %; flex sup = M. flexor digitorum superficialis; flex prof = M. flexor digitorum profundus; *PIP* proximal interphalangeal joint, *DIP* distal interphalangeal joint

13.3 Long-Term Damage

Whether rock climbing leads to degenerative arthritis of the finger joint has been debated and already investigated by various authors [1–5]. Radiographic adaptations and changes in long-term adult climbers are a well-known fact, and a positive correlation to their years of climbing has been shown [1, 2, 5–8, 21–23]. Many authors described radiographic changes of finger joints of long-term climbers such as osteophytes, subchondral sclerosis, and joint space narrowing [4]. Nevertheless, none of the previously mentioned authors could provide clear evidence of an increased rate of degenerative arthritis compared to an age-matched group of non-climbers [1–4]. Most of this research concerns adult long-term climbers (>5 years), whereas data on young climbers or the long-term effects of intensive climbing since childhood are still sparse [24].

For example, Rohrbough et al. [2] could not prove a significantly higher rate of osteoarthritis among climbers than non-climbers. These findings contradict a recent study by Allenspach et al.

[6]. They compared 31 male sport climbers to 67 non-climbers and found a significantly higher risk of osteoarthritis in the climbers ($p < 0.001$) [6]. While another study by Sylvester et al. [3] could not prove a higher risk of osteoarthritis in climbers' hands, it showed that climbers' fingers have an increased diameter with additional bone deposited subperiosteally. Significant predictors include the highest level achieved in bouldering and sport climbing [3]. Bollen et al. [1] and Hochholzer and Schöffl [5, 8, 21] found that 28–39% of climbers with more than 15 years of climbing experience showed osteoarthrotic findings on radiographs. However, these were mostly minor degenerative changes such as so-called mini-osteophytes in the PIP or MCP joints without any clinical relevance. Clear osteoarthroses could only be found in less than 10% of the climbers, whereas the climbers with condition after growth plate joint injuries stood out again (7%) (see Sect. 6.3.9). In contradiction to these radiographic findings, the respective climbers are sometimes surprisingly free of complaints [5] (Table 13.3).



Fig. 13.2 Lateral image of the ring finger of a male sport climber (age, 24 years; climbing time, 6 years). Distinct convex hypertrophy of the cortex leads to the narrowing of the medullary canal as an example of an adaptive reaction of the bone to the load during climbing



Fig. 13.3 Calcification of the tendon insertions of the superficialis, profundus, and extensor tendons in a climber after about 20 years of climbing

Table 13.3 Radiographic findings in 165 high-level climbers (second number in bracket in percent). The high number of osteoarthritic signs in climbers 5–10 years is due to epiphyseal fractures

Climbing years number	5–10 (n = 65)	>10 (n = 48)	>15 (n = 29)	>20 (n = 13)
No reaction	14 (22)	4 (8)	0 (0)	0 (0)
Stress reactions	51 (13)	44 (91)	29 (100)	13 (10)
Cortical hypertrophy	40 (37)	34 (70)	21 (72)	12 (92)
Subchondral sclerosis	28 (43)	27 (56)	23 (79)	12 (92)
Calcification of flexor tendons	12 (18)	16 (33)	8 (27)	7 (53)
Broadened joint base PIP	29 (45)	32 (66)	19 (65)	7 (53)
Broadened joint base DIP	37 (57)	37 (77)	26 (89)	8 (61)
Osteoarthrotic signs	11 (17)	20 (41)	18 (62)	4 (30)
Osteophytes PIP	3 (5)	11 (22)	17 (58)	4 (30)
Osteophytes DIP	0 (0)	4 (8)	4 (13)	4 (30)
Joint space narrowing	2 (3)	1 (2)	0 (0)	2 (15)
Incongruent joint space	6 (9)	1 (2)	0 (0)	3 (23)
Subchondral cysts	2 (3)	1 (2)	0 (0)	0 (0)



Fig. 13.4 Periarticular calcifications of the capsule and ligaments of a 35-year-old climber

Periarticular calcifications are relatively common, especially in the area of the ulnar and radial collateral ligament of the PIP joint. However, they rarely correlate with true joint space narrowing or other degenerative changes. They are usually due to poorly healed or unhealed injuries to the collateral ligaments. More often, however, the cause seems to be microinjuries of the capsule and ligaments in the PIP joint. When a “one finger pocket” is used in climbing, considerable torsional stress occurs, which stretches the capsular ligament apparatus in the PIP joints and causes stress (Fig. 13.4).

An increasing number of small osteophytes have been found on the MCP, PIP, and DIP joints in athletes who have been climbing for more than 10 years. These bone spurs are clearly visible in the a.p. image, mostly without changes in the lateral image. Clinically, these mini-osteophyte



Fig. 13.5 An a.p. X-ray of the ring finger of a male climber of the PIP joint (age, 32 years; climbing time, 12 years). Small osteophytes at the radial side of the PIP joint

formations did not show any symptoms or movement restrictions (Fig. 13.5).

However, it is extremely important to not only take a.p. X-rays but also lateral X-ray images. This is where the most serious degenerative changes become apparent in long-term climbers (>10 years). In 2010, we were able to examine a total of 13 climbers who showed pronounced dorsal and, less frequently, palmar osteophyte formation in the PIP and DIP joints. These mostly dorsal osteophyte (bone spur) formations are also called “extensor hood syndrome” [22] (see Chap. 6) (Table 13.4). Some of these climbers have impressive ossifications in the area of the capsule and extensor tendon apparatus. Clinically, a frequent additional observation to pressure pain was a slipping phenomenon in the area of the dorsal extensor apparatus at the PIP joint. This phenomenon could be regularly triggered by movement. The fact that the dorsal extensor hood and capsule are irritated in this case can be easily seen on the X-ray image. All of these climbers had a high climbing level and had already been climbing between 5 and 30 years (average 17.8 years); the average age was 32.5 years.

In addition to the dorsal and volar osteophytes, however, changes in the joint space were also evi-

Table 13.4 Description of climbers with extensor hood syndrome (Schöffl et al. [22])

No.	Side	Finger	Joint	Age	Climbing level	Climbing years	Kellgren score
1	Right	Middle	DIP	31	9.5	22	3
2	Right	Middle	PIP/DIP	44	10.3	30	4
3	Both	Middle	PIP	39	10.3	22	3 and 3
4	Both	Middle	PIP	42	10.3	21	3 and 4
5	Both	Middle	PIP	25	11.0	6	2 and 3
6	Both	Ring	PIP	37	11.3	17	2 and 2
7	Both	Middle	PIP	20	10.7	8	2 and 3
8	Right	Middle	DIP	33	10.0	23	3
9	Left	Ring	PIP	28	9.7	14	3
10	Right	Middle	DIP	43	10.3	18	3
11	Both	Middle	PIP	25	9.7	20	3 and 3
12	Both	Middle	PIP/DIP	55	9.0	30	3 and 4
13	Right	Ring	PIP	17	10.3	5	2

Climbing level UIAA metric, *DIP* distal interphalangeal joint; *PIP* proximal interphalangeal joint

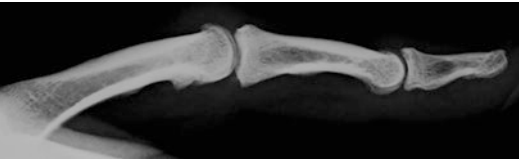


Fig. 13.6 Lateral X-ray of a long-time climber with dorsal and palmar osteophytes

dent, less with clear narrowing than more frequently with irregularities in the joint surfaces. After classification using the Kellgren-Lawrence classification, most of the changes in the score ranged between 2 and 4 (Fig. 13.6).

In contrast, fewer clinical complaints were reported at the DIP joint, despite the considerable formation of osteophytes. In all cases, there was a limitation of extension and flexion of up to 10°. These thickenings and swellings tend to be more painful for younger climbers, but the older climbers have probably learned to adapt the load to the pain (Figs. 13.7 and 13.8).

The importance of lateral X-ray views are also supported by a recent study by Allenspach et al. [6]. They investigated the anterior-posterior and lateral radiographic views of the fingers in a group of 31 high-level Swiss sport climbers. The mean years of climbing was 20 years, and the mean of the highest reached redpoint sport climbing level was 8b (UIAA 10) [6]. Almost all of the climbers had only little or no symptoms in the finger joints, but up to 84% showed osteophytes



Fig. 13.7 Widening of the DIP joints of a 55-year-old climber. Clinical limitation of movement, minor complaints

at the PIP-joints and up to 68% in the DIP-joints. According to the Kellgren and Lawrence score, six climbers (19%) had signs of an osteoarthritis (significant), whereas the age-matched non-climbing group had practically no signs of radio-

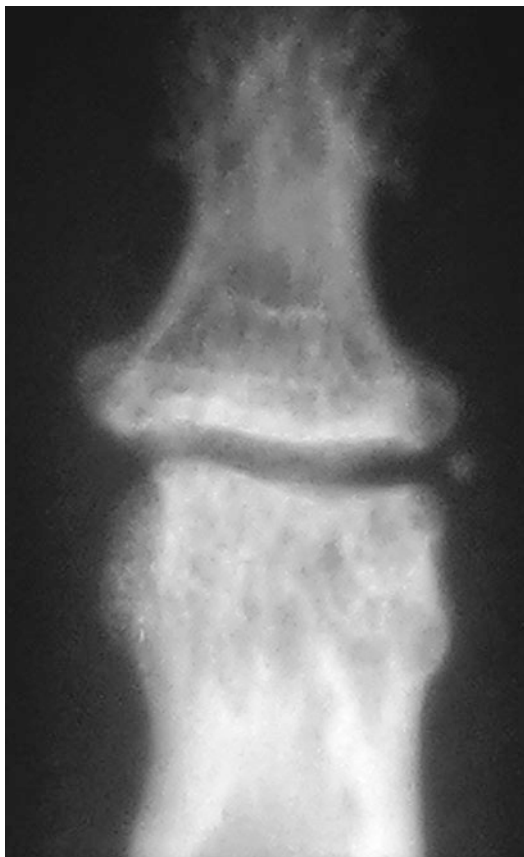


Fig. 13.8 X-ray of the DIP of the same climber: significant widening of the joint surfaces and small periarticular calcification

graphic changes in the finger joints. They observed that these bone spurs were best detected in the lateral radiographs [6]. These bone spurs can also be detected through ultrasound examination [25].

In comparison to known degenerative diseases of the hand (Bouchard's and Heberden's osteoarthritis), the absence of any inflammatory, destructive, or unstable joint changes is noticeable in the climber patients. The visible radiographic changes differ fundamentally from the images of known finger polyarthritis. There is also no climber known to us who has had to be fitted with a finger prosthesis.

In a recent study of the 11-year follow-up of our once young high-level climbers from the German Junior National Team, we found that one quarter of climbers who performed at a high

level in their youth showed a “mild” form of osteoarthritis (Kellgren II) at an age of less than 30 years [16]. During follow-up, 40% of the subjects of the GJNT and 23% of the RC group complained of swollen finger joints in the morning. Nevertheless, in all cases except one, the swelling was reported without any accompanied pain. Training intensity in 1999 ($p < 0.05$) and body weight in 1999 were significant ($p < 0.05$) for the development of radiographic stress reactions in 2011. Significant statistical influences for the development of early-onset osteoarthritis could be found for the overall sum of training years ($p = 0.024$), use of campus board training in 1999 ($p = 0.033$), and climbing level ($p = 0.030$).

13.4 Exemplary Cases

Three individual, exemplary cases show how the finger joints of long-time sport climbers react to the strain.

Climber A, born in 1951, started climbing at the age of 14 and climbed at a high level for decades (UIAA 10th grade (higher 5.13 YDS)). He was one of the pioneers for the introduction of the UIAA 7th grade (5.10 YDS) of difficulty in the alpine area. Between 1984 and 1987, the climber underwent surgeries on both hands for carpal tunnel syndrome and ulnar tunnel syndrome. For many years, the climber suffered from painful swelling of the fingers and a 15–20° extension deficit of all of the middle finger joints. The climber's semi-professional climbing career ended in 1994, when he was still suffering from significant pain and movement restrictions of the finger middle and end joints. In 1998, a radiographic examination was carried out on both hands. The finger joints were largely free to move and no longer showed any extension deficit; the climber no longer suffered from pain at rest or during exertion for recreational climbing tours in the lower difficulty range. Overall, only minor “mini-osteophytes” were found, particularly in the middle joints, and no destruction or narrowing of the joint space was found (Figs. 13.9, 13.10, and 13.11).



Fig. 13.9 Massively swollen middle finger joints at the time of active climbing



Fig. 13.10 MRI of the hands. The extension deficit in the PIP joints is clearly visible, as they are only partially depicted. Clearly hypertrophic small hand muscles, flexor tendons, and collateral ligaments. In the PIP joint, also minor effusion is evident. The MRI was performed in 1988

Climber S, born in 1965, started climbing at the age of 15. Soon afterwards, the level of difficulty of climbing increased to that of the top of the world with participation and achievements in



Fig. 13.11 X-ray image showing small capsule calcification in the DIP joint on the ring finger with significant widening of the end joint and less in the middle joint. Small, pointed osseous extension at the base joint. The joint spaces are wide and intact. X-rays were taken in 1998

World Championships and international competitions. Climber S established sensational first ascents on all continents. There were no major injuries to the hand or fingers and small wounds had always healed well. After 30 years of climbing, the X-rays of the hands are completely unremarkable (Fig. 13.12).

Climber K, born in 1954, started climbing at the age of 14. For decades, he climbed sport climbing routes and alpine routes up to the lower 10th UIAA grade (up to 5.13c) of difficulty and established many classic first ascents on all continents. In the course of his career, he sustained numerous injuries to fingers and hands, often including lengthy finger swellings, many of which probably did not heal properly. Climber K



Fig. 13.12 Hand a.p. at the age of 42 years. No degenerative changes. Slightly increased sclerosis of the joint surfaces with joint base widening

trained with one-finger pockets and hanging exercises on small edges for many years (Fig. 13.13).

13.5 Causes and Therapy

Many studies have shown that intensive and long-lasting sporting activity lead to reactions of the soft tissues and bone [26]. Leal et al. [27] were the first to report radiographic abnormalities in the fingers and hands of rock climbers. Heuck et al. [18] and Hochholzer et al. [19] demonstrated radiographic adaptations to the high stress of climbing using radiographic and MRI analyses. They found an adaptive hypertrophy of the joint capsule in the PIP and DIP joints, thickening of the collateral ligaments, cortical hypertrophy, and a hypertrophy of up to 50% of the flexor ten-



Fig. 13.13 PIP joint of a male climber (age, 54 years; climbing time, more than 30 years). The joint space is well preserved, even if it appears somewhat asymmetrical in the DIP joint and in the palmar part of the PIP joint. Noticeable large traction osteophytes at the PIP joint, which often irritate the extensor tendons and the capsule

dons themselves. They also showed that certain reactions, such as cortical hypertrophy, could be adaptive signs of the high impact to the fingers in climbing and could not be attributed to pathologic osteoarthritic changes. Similar findings of connective tissue adaptations were lately also reported by Schreiber et al. [14]. These reactions are probably due to subperiosteal bone formation. Hahn et al. [28] found an average 6% increase in cortical thickness and a 20% decrease in cancellous bone width in high-level sport climbers compared to a control group. Our study on young climbers of the German national climbing team shows how fast these reactions occur. After only 3 years of intensive climbing training, we found first significant cortical thickening and widening of the joint surfaces [15].

No major development of osteoarthritis is to be expected in sports with cyclical movements [29]. On the other hand, axial loads and high-pressure peaks in the joints under unfavorable joint positions, such as those frequently encountered in climbing, are very stressful for the joints [26, 30, 31]. Osteoarthrotic conditions in long-term climbers are well-known, and several studies have shown that osteoarthrotic changes in the hands of rock climbers increase with the number of years and the level of climbing [1–3, 6, 8, 9, 14, 28].

The development of finger polyarthrosis in sport climbers is certainly multifactorial and depends on many different exogenous and individual genetic factors. The genetically deter-

mined connective tissue structure and possible hypermobility influence the stress on joints and thus the development of osteoarthritis.

In the field of sports, the most common cause for the development of degenerative diseases is probably overloading with microinjuries. Epiphyseal fractures in young climbers can lead to an early onset of osteoarthritis [32–34].

The crimping finger position with hyperflexion in the middle joint and a final extension in the DIP joint probably leads to microinjuries in the sensitive cartilage area [18]. There seems to be evidence that climbers show abnormal radiographic findings that have been underestimated or unhealed overload reactions with joint swelling. There may also be reason to discuss that some forms of training, such as unusually intensive use of the smallest crimps and the campus board, are more likely to cause degenerative changes. Our recent longitudinal study on the onset of osteoarthritis in young rock climbers found significant statistical influences on the development of early-onset osteoarthritis from the overall sum of training years ($p = 0.024$), use of campus board training in 1999 ($p = 0.033$), and the climbing level ($p = 0.030$) [16]. One quarter of climbers who performed at a high level in their youth showed a “mild” form of osteoarthritis (Kellgren II), at ages of under 30. Following the training regimes of the climbers for more than 10 years showed that intensive finger training (e.g., campus board training) can lead to early-onset osteoarthritis of the hand.

Therefore, the most sensible step to avoid degenerative changes of the finger joints seems to be the optimization of the load during training. This includes the avoidance of overstrain reactions with swelling of the finger joints, sensible training schedules with conscious recovery phases, and the complete healing of injuries and overstrain reactions. Stressful forms of training involving with hanging exercises on very narrow crimps or intensive training on campus boards seem to increase the overload reactions of the finger joints.

After intensive training, it is certainly advisable to “post-treat” the finger joints, for example, with hand putty, small bags filled with sand, or

acupressure rings. Concentric movements of the joints without force peaks promote blood circulation and regeneration.

If complaints occur, the first measure should be a rest from climbing, must be taken until the complaints have subsided. The joint effusions of the PIP joints should always be taken and treated first. In addition to ice treatments of all kinds, topical anti-inflammatory ointments, short-term oral NSAID administration, and minimal injections of a corticoid with a corresponding break of 2–3 weeks can result in improvement in therapy-resistant cases. Chondroitin and glucosamine seem to modulate the symptoms in the long term. In the case of symptomatic complaints of the extensor hood, a single targeted injection at the point of pain in the PIP joint with a highly diluted corticosteroid can often bring long-term improvement. In few cases, surgical removal of the traction osteophytes was necessary, yielding mostly good results. Radial shockwave therapy and hand baths with anti-inflammatory sulfur solutions, as used in rheumatoid arthritis patients, also help.

In the overview of the X-rays, especially among climbers who have been practicing this sport for a long time (longer than 10 years), it is astonishing that there appear to be relatively few severe degenerative changes. Only a few climbers were found to have slightly incongruent joint spaces at the PIP and DIP joints, without the joint space being massively narrowed. It is interesting to note that the clinical changes with movement restrictions often regressed after months to years, and the radiologically minor changes were clinically insignificant.

Another interesting aspect is that some of the long-time high-level climbers display no degenerative changes at all. However, the extent to which a very early start with climbing in childhood or adolescence influences the onset of early finger arthrosis remains to be seen.

13.6 Summary

In summary, it can be said that many years of climbing in the high difficulty range leads to increased degenerative changes in the middle and

end joints of the fingers. This can be seen radiologically in about 30% of patients who have climbed for more than 15 years. Clinically, even very significant changes are surprisingly rarely accompanied by serious impairments. Individual factors, intensity of climbing, number of climbing years, injuries, and the type of training methods seem to be decisive for the development of finger joint arthrosis.

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